

Fourier Transform and Spectral Derivatives

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1 Introduction to the Fourier Transform

1.1 The main idea and the areas of application

In the entirety of Physics, Mathematics and Engineering very few topics appear frequently. These are topics that appear at every stage, as you go deeper into the subject, with their meaning and math changing and developing—becoming more refined. At every level they hold a different meaning and philosophy. The Fourier Transform is one such mathematical object which keeps appearing in the literature of Science and Engineering all the time: be it Electrical Circuits, Heat flow, Wave mechanics, Optics, Electromagnetism, Quantum Mechanics and even Quantum Computing. The list goes on. But what exactly is the Fourier Transform?

The main idea of the Fourier Transform is to take a signal as an input and decompose it into the various pure single-frequency signals in it. That's it! But you can use it to do tasks that seem remotely possible. For example: Solving differential equations, or Image compression!

A list of areas where Fourier Transform is used is:

1. Signal Processing (filtering, compression, spectral analysis)
2. Image Processing (denoising, edge detection, image compression)
3. Quantum Mechanics (wavefunctions, momentum-position duality)
4. Quantum Computing (Shor's Algorithm, Quantum Fourier Transform)
5. Partial Differential Equations (heat equation, wave equation, Schrödinger equation)
6. Communications Engineering (modulation, bandwidth analysis)
7. Audio and Speech Analysis (pitch detection, sound synthesis)
8. Medical Imaging (MRI, CT reconstruction)
9. Optics and Diffraction Theory (Fraunhofer diffraction, lens systems)
10. Seismology and Geophysics (earthquake wave analysis)
11. Machine Learning and Data Science (feature extraction, frequency-domain analysis)

1.2 History of the Transform

In 1822, Joseph Fourier claimed that any function, whether continuous or discontinuous can be expanded into “a series of sines”. Fourier Transform was not the way we know it today. It was primitive. That important work was corrected and expanded upon by others to provide the foundation for the various forms of the Fourier transform used since.

Whom to credit for coming up with the idea of the Fourier transform is still a matter of debate. The idea of using a harmonic series to study astronomical positions was present in the ancient Babylonians and Greeks.

In modern times, variants of discrete Fourier transform were used by French mathematician, astronomer and physicist Alexis Clairaut, in 1754, to compute an orbit.

In 1759, Joseph Louis Lagrange used it for vibrations on a string.

Carl Friedrich Gauss, in 1805, was the first person to use both—sine and cosine terms for DFT.

The first fast Fourier transform (FFT) algorithm for the DFT was also discovered around 1805 by Gauss when interpolating measurements of the orbit of the asteroids Juno and Pallas.

The modern day discovery of the FFT algorithm came as a necessity to optimise the DFT, in order to read ground signals to check out for nuclear weapon testing. The algorithm, also known as Cooley-Tukey Algorithm, was mathematically formulated by American mathematician J.W. Cooley and the program was developed by American mathematician and statistician John Tukey. We will be mostly interested in the radix-2 version of it.

2 Special Functions, L^2 spaces, Fourier Series & Transform

2.1 The Dirac delta Function

Named after Physicist Paul Dirac, the Dirac delta function (or δ distribution), also known as the unit impulse, is a generalized function on the real numbers, whose value is zero everywhere except at zero, where it is infinite, and whose integral over the entire real line is equal to one.

$$\delta(x) = \begin{cases} 0 & x \neq 0 \\ \infty & x = 0 \end{cases}$$
$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

Rather than the exact definition of this function it is a lot more useful to see how it works with functions:

$$\int_{-\infty}^{\infty} f(x) \delta(x - a) dx = f(a)$$

whenever $f(a)$ is defined. An easy intuitive visualisation of the function is to imagine a rectangle of fixed unit area lying along its breadth on the X axis. Let the breadth of the rectangle shrink to infinitesimal dimensions. That's the Dirac delta function. When the idea of the function was first conceived, there was a huge debate as to whether it was truly a "function" in the mathematical sense. However, the function was really helpful in modeling impulses, and studying discrete mass and charge systems in physics. Now it is included in the class of generalised functions.

We can go for a not-so-rigorous, intuitive proof of the above-mentioned properties of the Dirac delta function. Consider the discretisation of the X-axis into steps of length Δx . In such a case with the rectangular approximation of our Dirac delta function, the closest we reach is

$$\delta_{\text{Rect}}(k) = \begin{cases} 0 & k \neq 0 \\ \frac{1}{\Delta x} & k = 0 \end{cases}$$

such that $x = k\Delta x$ for some integer k . The approximation is equal to the actual case when $\Delta x \rightarrow 0$.

Using this new definition,

$$\begin{aligned} \int_{-\infty}^{\infty} \delta(x) dx &\approx \sum_{k=-\infty}^{\infty} \delta_{\text{Rect}}(k) \Delta x \\ &= \delta_{\text{Rect}}(0) \frac{1}{\Delta x} \\ &= 1 \end{aligned}$$

and, similarly you prove the latter equation. There is one more definition of this function which is genuinely useful, the one we are going to use more often and it is:

$$\delta(x) = \int_{-\infty}^{\infty} e^{2i\pi kx} dk$$

Let us sit for a moment and try to understand why it is true. Well, the integral

$$\begin{aligned} I_N(x) &= \int_{-N}^N e^{2i\pi kx} dk \quad \text{for } N \in \mathbb{Z} \\ &= \left(\frac{e^{2i\pi kx}}{2i\pi x} \right)_{-N}^N \\ &= \frac{\sin(2N\pi x)}{\pi x} \end{aligned}$$

Now,

$$\lim_{x \rightarrow 0} I_N(x) = 2N$$

For the continuity of the function we define $I_N(0) = 2N$. Evidently $I_N(x) = 2N\text{sinc}(2N\pi x)$. So, now to prove our claim we need to show

$$\lim_{N \rightarrow \infty} I_N(x) = \delta(x)$$

How do we prove that? Well we had shown two key properties for the Dirac delta function. We need to just prove that our $I_N(x)$ in the limiting case of $N \rightarrow \infty$ satisfies those properties. To prove the first property:

$$\lim_{N \rightarrow \infty} I_N(x) = \lim_{N \rightarrow \infty} \frac{\sin(2N\pi x)}{\pi x} \quad \text{when } x \neq 0$$

But hey! there is no pointwise limit in this case. Obviously the above limit is bounded between $-\frac{1}{\pi x}$ and $\frac{1}{\pi x}$ but it keeps on oscillating. The complete proof of this is beyond the scope of what we are doing here (typically it involves the theory of distributions) and so, we shall rather ignore this case. Moving on, for the $x = 0$ case it is rather easy to prove $I_N(0) \rightarrow \infty$ as $N \rightarrow \infty$.

To prove the second property,

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{\sin(2N\pi x)}{\pi x} dx &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin(t)}{t} dt \\ &= 1 \end{aligned}$$

which proves our claim.

2.2 The Spaces that we live in: Vector Spaces, Inner Product Spaces, Hilbert Spaces and L^2 Spaces

We shall take some time to develop the notion of some spaces.

2.2.1 Vector Spaces

A vector space is a concept that one first encounters while learning linear algebra.

Formally, a **vector space** V over a field F is a non-empty set together with

- **Vector addition**

$$+ : V \times V \rightarrow V, \quad (v, w) \mapsto v + w,$$

where $v, w \in V$.

- **Scalar multiplication**

$$\cdot : F \times V \rightarrow V, \quad (c, v) \mapsto c \cdot v,$$

where $c \in F$ and $v \in V$.

These operations satisfy the following axioms for all $u, v, w \in V$ and all $a, b \in F$:

1. **Commutativity of addition**

$$u + v = v + u.$$

2. **Associativity of addition**

$$(u + v) + w = u + (v + w).$$

3. **Additive identity**

There exists an element $0 \in V$ such that

$$v + 0 = v$$

for every $v \in V$.

4. **Additive inverse**

For every $v \in V$, there exists an element $-v \in V$ such that

$$v + (-v) = 0.$$

5. **Multiplicative identity**

$$1v = v,$$

where 1 is the multiplicative identity of the field F .

6. Distributive properties

$$\begin{aligned}a(u + v) &= au + av, \\(a + b)v &= av + bv.\end{aligned}$$

7. Compatibility of scalar multiplication

$$a(bv) = (ab)v.$$

Given we know this, what could the elements of a vector space be? Well, any set that satisfies this condition is a vector space. Common examples of vector spaces are: $\mathbb{R}^n$ (over $\mathbb{R}$), $\mathbb{C}^n$ (over $\mathbb{R}$ or $\mathbb{C}$), $M_{n \times n}(\mathbb{R})$, etc. There are a lot more technicalities of Vector spaces and linear algebra but that needs a separate discussion of its own. Here we shall bypass all that discussion, since we do not need all of that. However we will give some time to discuss the basis. But after we have defined inner-products and norms.

2.2.2 Inner Product Spaces

From high-school mathematics a lot of us are familiar with the idea of a “dot product” also known as an inner product. From hereby we try to abstractify the definition.

An inner product space is a vector space V over a field F endowed with an inner product, that is a map

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow F$$

This inner product has some special properties. For $x, y, z \in V$ and $a, b \in F$

1. Conjugate Symmetry

$$\langle x, y \rangle = \overline{\langle y, x \rangle}.$$

2. Linearity in the first argument

$$\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle.$$

3. Positive Definiteness

$$\langle x, x \rangle \geq 0, \quad \langle x, x \rangle = 0 \iff x = 0.$$

Also this helps us define the norm of a vector (the length) as $\|x\| = \sqrt{\langle x, x \rangle}$.

Quick question: What is the guarantee that we are ensuring that the length is always real? The proof is really simple. It is due to positive definiteness. Inner products and inner product spaces can have a lot more interesting properties exploiting the above properties. Without going deep into all the technicalities, we move on to some special inner product spaces.

1. $\mathbb{R}^3$

Although we could deal with $\mathbb{R}^n$ in general, I chose this inner product space as this is the most familiar one. The elements or “vectors” of this space can be denoted by many notations: The column vector notation (3×1 matrix), the row vector notation (1×3 matrix), the 3-tuple notation, and using the unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$. Of these, I choose the 3-tuple notation.

For $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$, where $u_1, u_2, u_3, v_1, v_2, v_3 \in \mathbb{R}$, the inner product is defined as $\langle u, v \rangle = u_1v_1 + u_2v_2 + u_3v_3$. This simple definition is also known as the dot product. And it satisfies all the properties. For the general case, if $u, v \in \mathbb{R}^n$, then $u = (u_1, \dots, u_n)$ and $v = (v_1, \dots, v_n)$,

$$\langle u, v \rangle = \sum_{k=1}^n u_k v_k$$

Interestingly, we can extend the definition of an “angle” or “length” from 2-D or 3-D space to n-D space. Since $\langle u, v \rangle = \|u\| \|v\| \cos \theta$, where θ is the angle between the two vectors. Then

$$\begin{aligned}\cos \theta &= \frac{\sum_{k=1}^n u_k v_k}{\|u\| \|v\|} \\&= \frac{\sum_{k=1}^n u_k v_k}{(\sum_{k=1}^n u_k^2)^{\frac{1}{2}} (\sum_{k=1}^n v_k^2)^{\frac{1}{2}}}\end{aligned}$$

In the next part about functions we’ll see how this idea of angles carry over, and how two functions could be perpendicular! By the way if two vectors are perpendicular then $\langle u, v \rangle = 0$

2. Functions

Since we are moving from the discrete domain to the continuous domain, it is noteworthy to mention that the summation of all the components changes into an integral over a continuous domain. Which domain? Well since lengths are always finite, wherever the integral converges (or the functions are defined) we choose that as the domain. So,

$$\langle f, g \rangle = \int_D f(x) \overline{g(x)} dx$$

D could be $(-\pi, \pi)$, $(0, 1)$, $(-1, 1)$ or even $\mathbb{R} = (-\infty, \infty)$. A good check will be to see that all this definition satisfies all the given properties.

2.2.3 Hilbert Spaces

A Hilbert space is an inner product space that is complete with respect to the norm induced by its inner product. A Cauchy sequence a_n is a sequence whose terms come arbitrarily close together, as n gets very large. Such a sequence will always have a limit. But will that limit exist in the same space as the terms? Consider the set of rational numbers Q . Let us define a sequence:

$$\begin{aligned} u_0 &= 3 \\ u_1 &= 3.1 \\ u_2 &= 3.14 \\ u_3 &= 3.141 \\ u_4 &= 3.1415 \end{aligned}$$

and so on. This is a very frequently cited example to show “completeness” of a space. The limit of this sequence is π but $\pi \notin Q$. So Q is not complete. However if we considered $\mathbb{R}$, we would have a complete set. Further, allowing an inner product (say the conventional dot product, which for $\mathbb{R}$ would be multiplication) together with this, we get a Hilbert Space.

Consider another example:

$$p_n(x) = \sum_{k=0}^n \frac{x^k}{k!}$$

is an element of $P = \{a_0 + a_1x + \dots + a_nx^n : n \in \mathbb{N}, a_i \in \mathbb{R}, i = 1, 2, 3, \dots\}$. Considering these polynomials on $[0, 1]$ and endowing them with the inner product $\langle p, q \rangle = \int_0^1 p(x)q(x) dx$, it is easy to show that this is an inner product space. However, $\lim_{n \rightarrow \infty} p_n(x) = e^x \notin P$ and so P is not a Hilbert space. So how do we define a space for functions, that is a Hilbert Space all the way? Enter L^2 spaces.

2.2.4 L^2 Spaces

Well although L^2 Spaces are not the only way to solve the problem, they are one of the most common ways, and honestly? This is what we’ve actually been building towards for so long, but had we just dropped this in the beginning, we’d have left a lot of questions unanswered. Although a lot of texts either assume the reader is familiar with the previous concepts or decide to totally ignore it.

Recall the euclidean norm. In $\mathbb{R}^n$ the “distance” between two points in the space (elements of the set) P and Q is given by

$$\begin{aligned} \|P - Q\| &= \sqrt{\langle P - Q, P - Q \rangle} \\ &= \sqrt{\sum_{k=1}^n (p_k - q_k)^2} \end{aligned}$$

where $P = (p_1, \dots, p_n)$ and $Q = (q_1, \dots, q_n)$. So for functions let $D \subseteq \mathbb{R}$ so $L^2(D)$ is the Hilbert space of functions over D which have a finite norm.

And this space is so very important — the wavefunction of Quantum Mechanics that the wavefunction is a valid

one (i.e., solves the Schrodinger's Equation) only if it exists in $L^2(R)$.
A function f is in $L^2(D)$ if

$$\int_D |f(s)|^2 ds < \infty$$

The Born Rule in Quantum Mechanics states: The square modulus of the wavefunction represents the probability density. So in position space form, in one dimension:

$$\begin{aligned}\frac{dP}{dx} &= \psi(x)\overline{\psi(x)} = |\psi(x)|^2 \\ \int_{-\infty}^{\infty} dP &= 1 \\ \int_{-\infty}^{\infty} |\psi(x)|^2 dx &= 1\end{aligned}$$

Every vector space has linearly independent sets that span the entire space. These are known as the basis of the space. For example the standard basis of R^2 is $\{(1,0), (0,1)\}$. Any vector $v \in R^2$ can be expressed as a linear combination of the basis vectors: $v = x(1,0) + y(0,1) = (x,y)$. But the thing about the basis is there are infinitely many such bases. We are specifically interested in orthonormal basis vectors:

A finite basis B is orthonormal iff:

$$\begin{aligned}B &= \{v_1, v_2, v_3, \dots, v_n\} \\ \langle v_i, v_j \rangle &= \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}\end{aligned}$$

or, in other words, $\langle v_i, v_j \rangle = \delta_{ij}$.

For a Hilbert Space (which is also a Vector Space) like $L^2([0,1])$, the basis is infinite dimensional (simply put, infinite cardinality of basis), and is

$$B = \{e^{2\pi i x n} : n \in Z\}$$

where $v_n(x) = e^{2\pi i n x}$ is a function. Check the inner product:

$$\begin{aligned}\langle v_n(x), v_m(x) \rangle &= \int_0^1 v_n(x) \overline{v_m(x)} dx \\ &= \int_0^1 e^{2\pi i(n-m)x} dx \\ &= \begin{cases} 0 & m \neq n \\ 1 & m = n \end{cases}\end{aligned}$$

For some space like $L^2([-N, N])$, we have $B = \{\frac{e^{2\pi i n x}}{\sqrt{2N}} : n \in Z\}$

It is important to realise that for $L^2(R)$ these basis vectors cannot be used since these are not in $L^2(R)$! To show that:

$$\int_{-\infty}^{\infty} e^{2\pi i n x} \overline{e^{2\pi i n x}} dx = \int_{-\infty}^{\infty} dx = \text{diverges}$$

However we shall keep using this basis, and we will not discuss as to why we are doing so, because the justification is beyond the scope of the text. The reader shall find that the things still “work”, as it would if these were the basis.

2.3 The Fourier Series and Transform

2.3.1 The Fourier Series

The idea of the Fourier Series as stated earlier, was to write it out as a sum of pure signals — a linear combination of them. Hence, we could have taken cos or sin terms. But due to Euler's formula $e^{2\pi i n x} = \cos(2\pi n x) + i \sin(2\pi n x)$

we can do both the terms together using one term. Also when we go for spectral derivatives we'll see how this benefits us hugely. Let us take a sample signal (function) f for example. To decompose it into several discrete frequencies, let the corresponding weight of the frequency n be $\hat{f}(n)$. So the function decomposes as

$$f(x) = \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{2\pi i n x}$$

There are quite a lot of important features that come up with this decomposition.

$$\begin{aligned} 1. \quad f(x+1) &= \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{2\pi i n (x+1)} \\ &= \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{2\pi i n x} e^{2\pi i n} \\ &= f(x) \\ 2. \quad \overline{f(x)} &= \sum_{n=-\infty}^{\infty} \overline{\hat{f}(n)} e^{-2\pi i n x} \\ f(x) \overline{f(x)} &= \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{2\pi i n x} \sum_{m=-\infty}^{\infty} \overline{\hat{f}(m)} e^{-2\pi i m x} \\ |f(x)|^2 &= \sum_{m,n \in \mathbb{Z}} \hat{f}(n) \overline{\hat{f}(m)} e^{2\pi i (n-m)x} \\ \int_0^1 |f(x)|^2 dx &= \int_0^1 \sum_{m,n \in \mathbb{Z}} \hat{f}(n) \overline{\hat{f}(m)} e^{2\pi i (n-m)x} dx \\ &= \sum_{m,n \in \mathbb{Z}} \hat{f}(n) \overline{\hat{f}(m)} \int_0^1 e^{2\pi i (n-m)x} dx \\ &= \sum_{m,n \in \mathbb{Z}} \hat{f}(n) \overline{\hat{f}(m)} \delta_{mn} \\ &= \sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2 \end{aligned}$$

Point number 1. says that the function f is 1-periodic. That is, it has periodicity 1. But how is that possible? We had chosen any random function f and there is no way that it certainly has a periodicity 1. The truth is, the term of the Fourier series is an approximation for the domain $(0, 1)$ (because the basis vectors were chosen so). This is important to mention. You can indeed get a better approximation by modifying the basis, for example, using $\frac{e^{i\alpha x}}{\sqrt{2\pi}}$ instead of $e^{2\pi i n x}$ we would have the periodicity 2π (also note the corresponding normalisation factor to maintain orthonormality, not only orthogonality).

Point number 2. states an important theorem known as Parseval's theorem (one of the many forms of the theorem). It states that the total spectrum (sum of the square modulus of the Fourier coefficients) is equal to the $L^2([0, 1])$ norm of the signal.

One last thing before we move on to the transform: How can we determine the Fourier coefficients? The answer could be just put forth as an integral but there is a very elegant aspect to it.

Remember how these are actually vectors? And these complex exponentials are “orthonormal basis” or in a High School physics way — unit vectors? We utilise that property. Take the inner product of the function f with a complex exponential to determine the “projection along that direction” or the corresponding Fourier coefficient.

$$\begin{aligned} \hat{f}(n) &= \langle f, e^{2\pi i n x} \rangle \\ &= \int_0^1 f(x) e^{-2\pi i n x} dx \end{aligned}$$

2.3.2 The Fourier Transform

With the Fourier transform the question becomes more direct — now we shall lose a bit of our innocence, that is, we are not looking for answers as to “what frequencies does this signal contain?” We want to know:

Given a function f which lives in time or space domain, what is the corresponding function $\hat{f}$ that lives in the frequency domain that gives the weights of the constituent signals in it.

Here's a subtlety: this weight can be a complex number that encodes the amplitude and phase information of the signal. Let us now directly go to mathematically define the Fourier Transform:

$$\hat{f}(s) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i s x} dx$$

Look at the $e^{-2\pi i s x}$ term. Whenever I look at it, I always feel that when it is multiplied with f and integrated over the entire real line, it has the effect of “uncoiling” the frequencies to find the specific frequency. Although it could be debated whether this visualisation helps the reader or not. Even if it doesn't we could clearly investigate this more rigorously so that we have an idea of what is actually going on in that formula. (Don't worry: Uncoiling is not an actual mathematical thing! ... at least in this case.).

I'd like to spend some time exploring the formula. Since the complex exponentials act as “unit vectors”, $\hat{f}(s)$ should also be a vector. We could show that using the following:

$$\begin{aligned}\overline{\hat{f}(s)} &= \int_{-\infty}^{\infty} \overline{f(y)} e^{2\pi i s y} dy \\ \hat{f}(s) \overline{\hat{f}(s)} &= \int_{-\infty}^{\infty} f(x) e^{-2\pi i s x} dx \int_{-\infty}^{\infty} \overline{f(y)} e^{2\pi i s y} dy \\ |\hat{f}(s)|^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x) \overline{f(y)} e^{2\pi i (x-y)s} dx dy \\ \int_{-\infty}^{\infty} |\hat{f}(s)|^2 ds &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x) \overline{f(y)} e^{2\pi i (x-y)s} dx dy ds\end{aligned}$$

We have arrived here at a very crucial step. $\int_{-\infty}^{\infty} |\hat{f}(s)|^2 ds = \|\hat{f}(s)\|^2$ as according to $L^2(R)$ norm. Now we are doing a rearrangement. And we'll use the dirac delta function properties that we talked about earlier.

$$\begin{aligned}\|\hat{f}(s)\|^2 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x) \overline{f(y)} \left(\int_{-\infty}^{\infty} e^{2\pi i (x-y)s} ds \right) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x) \overline{f(y)} \delta(x-y) dx dy \\ &= \int_{-\infty}^{\infty} f(x) \left(\int_{-\infty}^{\infty} \overline{f(y)} \delta(x-y) dy \right) dx \\ &= \int_{-\infty}^{\infty} f(x) \overline{f(x)} dx \\ &= \int_{-\infty}^{\infty} |f(x)|^2 dx \\ &= \|f(x)\|^2\end{aligned}$$

Since norm is always positive,

$$\|\hat{f}\| = \|f\|$$

This is a beautiful result, since it states that the norm of the Fourier transform of a function is the same as that of the function. Obviously it is true only if the norm is finite and so we recall one important point: These functions should lie in $L^2(R)$.

Look at the formula again. It is kind of changing the Fourier Series into a kind of continuum, instead of using the discrete frequency bins we had while using the integer indexing.

Also we do not have the transform having any inherent periodicity like we had for Fourier Series.

Moving on to the next question: If we have transformed our function to the frequency domain, can we bring it back to the space / time domain? In other words, can we have an inverse Fourier transform? We start with the same

equation:

$$\begin{aligned}
\hat{f}(s) &= \int_{-\infty}^{\infty} f(y) e^{-2\pi i s y} dy \\
\hat{f}(s) e^{2\pi i s x} &= e^{2\pi i s x} \int_{-\infty}^{\infty} f(y) e^{-2\pi i s y} dy \\
&= \int_{-\infty}^{\infty} f(y) e^{2\pi i (x-y)s} dy \\
\int_{-\infty}^{\infty} \hat{f}(s) e^{2\pi i s x} ds &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(y) e^{2\pi i (x-y)s} dy ds \\
&= \int_{-\infty}^{\infty} f(y) \left(\int_{-\infty}^{\infty} e^{2\pi i (x-y)s} ds \right) dy \\
&= \int_{-\infty}^{\infty} f(y) \delta(x-y) dy \\
&= f(x)
\end{aligned}$$

See, now the same derivation can be produced with inner-product argument.

The vector, or function $f(x)$ has projection $\hat{f}(s)$ when it is projected along $e^{2\pi i s x}$. Why? Because when we consider the $L^2(R)$ inner product,

$$\langle f(x), e^{2\pi i s x} \rangle = \int_{-\infty}^{\infty} f(x) e^{-2\pi i s x} dx = \hat{f}(s)$$

When we try to reconstruct f from the projections, we do this:

$$\begin{aligned}
f(x) &= \int_{-\infty}^{\infty} \langle f(x), e^{2\pi i s x} \rangle e^{2\pi i s x} ds \\
&= \int_{-\infty}^{\infty} \hat{f}(s) e^{2\pi i s x} ds
\end{aligned}$$

There is however a problem with this proof. Why? Because the complex exponentials need to be unit vectors, that is, have a unit norm. But... that is not the case — at least we have not discussed the validity of it. So I'd prefer the previous proof over this.

But most functions are not in $L^2(R)$. For example, I am interested in the Fourier transform of $f(x) = x$. So, how do I? We can do a trick — we can limit the domain to a finite interval. Let us have a domain $[-N, N]$ for $N \in \mathbb{N}$. The Fourier transform then is just an approximation. Also orthonormal basis on this domain will have the normalisation factor $\frac{1}{\sqrt{2N}}$. So, let F denote the Fourier functional. Then,

$$\begin{aligned}
F\{x\}(s) &= \frac{1}{\sqrt{2N}} \int_{-N}^N x e^{-2\pi i s x} dx \\
&= \frac{i}{\sqrt{2N}} \int_{-N}^N x \sin(2\pi s x) dx \\
&= i \sqrt{\frac{2}{N}} \int_0^N x \sin(2\pi s x) dx \\
&= i \sqrt{\frac{2}{N}} \left(\int_0^N x \sin(2\pi s x) dx \right) \\
&= \frac{i}{2\pi^2 s^2 \sqrt{2N}} (1 + 4\pi^2 s^2 N^2)^{\frac{1}{2}} \sin(2\pi s N + \phi_N(s))
\end{aligned}$$

where $\sin \phi_N(s) = \frac{1}{2\pi s \sqrt{1+N^2}}$ and $\cos \phi_N(s) = \frac{-N}{\sqrt{1+N^2}}$. For a large enough N , this simplifies to

$$\begin{aligned}
F\{x\}(s) &\approx -i \sqrt{\frac{N}{2}} \frac{\sin(2\pi s N)}{\pi s} \\
\lim_{s \rightarrow 0} F\{x\}(s) &= -i \sqrt{2N}^{\frac{3}{2}}
\end{aligned}$$

Clearly this transform explodes with increasing N . Also this does not look very friendly. Moreover, we have taken a very sweet symmetric domain with endpoints on natural numbers. Also not all functions have a closed form solution in the process of integration. Further, a lot of times we are not given a function: we are given a set of data to take a Fourier transform, for example in cases of compression the data is discrete. There is almost no real world application of this classical Fourier transform. So we have to develop some method to take the Fourier transform of a discrete set of points.

3 Discretisation: DFT and FFT

3.1 Why do we need to discretise?

I have already stated in the last paragraph of the previous section as to why we should have a discrete version of the transform. But, Fourier transform was not the first necessity to create a discrete algorithm. Computers work in discrete data points. It cannot integrate. It has to be programmed to do numerical integration. So, we have discrete approximations of important algorithms, like the Fourier transform.

3.2 The Discrete Fourier Transform

Consider that you have a function f . You have a domain $[a, b)$ such that $b - a = L$ and step size h such that number of points $N = \frac{b-a}{h}$. So we have the equation $Nh = L$. So the points are like $a, a + h, a + 2h, \dots, a + (N - 1)h$. We could have a discrete variable $x_k = a + kh$, with $k = 0, 1, 2, \dots, N - 1$. Now consider the original formula:

$$\hat{f}(s) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i s x} dx$$

We can convert this into a summation (Riemann Summation):

$$\hat{f}(s) \approx \sum_{k=0}^{N-1} f(x_k) e^{-2\pi i s x_k} h$$

At first glance, this appears to solve our problem. Given the sampled values of f , we can approximately compute $\hat{f}(s)$. However, a subtle issue remains. The variable s is still continuous, meaning that the above expression can be evaluated at infinitely many frequencies. Since our data consists of only N samples, it is unreasonable to expect it to determine infinitely many independent frequency components. This suggests that the frequency variable itself should also be discretised.

How do I discretise s ? Before we move to the formula, let's see the motivation for this. Recall that the length of the interval for which we have the signal values is L . Consider this as a signal on a time domain, let's say a sound wave. What are the frequencies you could fit in this interval?

For simplicity, consider a pressure wave $p(x, t) = p_0 \sin(\frac{2\pi}{\lambda} x - 2\pi s t)$, where λ is the wavelength and s is the frequency. The magnitude of the velocity of the wave is $v = \lambda s$. We could use complex numbers to represent this: $p(x, t) = p_0 e^{2\pi i (\frac{x}{\lambda} - s t)}$ or $\phi(t) = A e^{-2\pi i s t}$ where $A = p_0 e^{2\pi i x / \lambda}$ for some constant x . If a wave has a frequency s , the time period would be $\frac{1}{s}$.

Given we have a signal over a domain length L since the length is finite, we could approximate it to be L -periodic. This is actually the most intelligent choice and not at all random. This is because since we have the information about the function of a domain length L , for the rest of the domain instead of adding more information to the function that will make our problem much more difficult we make it periodic. Thus analysing over a window of length L will suffice than the entire real line. So, $\phi(a) = \phi(a + L)$ or, $2\pi s L = 2\pi n$ for some integer n . We could then discretise this variable:

$$s_n = \frac{n}{L}$$

So we have frequencies $0, \frac{1}{L}, \frac{2}{L}, \frac{3}{L}, \dots, \frac{N-1}{L}$. So our formula becomes

$$\begin{aligned}
\hat{f}(s_n) &= \sum_{k=0}^{N-1} f(x_k) e^{-2\pi i s_n x_k h} \\
&= \sum_{k=0}^{N-1} f(x_k) e^{-2\pi i \frac{n}{L} (a+kh)} h \\
&= \sum_{k=0}^{N-1} f(x_k) e^{-2\pi i a n/L} e^{-2\pi i k h n/L} h \\
&= h e^{-2\pi i a n/L} \sum_{k=0}^{N-1} f(x_k) e^{-\frac{2\pi i k n}{N}}
\end{aligned}$$

The prefactor could be set independent of n by choosing $a = 0$ without any loss of generality, and we can ignore the h term, and so, $F_n = \frac{\hat{f}(s_n)}{h}$, and $f_k = f(x_k)$ the formula becomes

$$F_n = \sum_{k=0}^{N-1} f_k e^{-\frac{2\pi i k n}{N}}$$

This is the discrete Fourier transform. So we have $F_0, F_1, \dots, F_{N-1}$. We can write this in a vector form. So we define

$$\mathbf{F} = \begin{bmatrix} F_0 \\ F_1 \\ F_2 \\ \vdots \\ F_{N-1} \end{bmatrix}$$

and

$$\mathbf{f} = \begin{bmatrix} f_0 \\ f_1 \\ f_2 \\ \vdots \\ f_{N-1} \end{bmatrix}$$

So,

$$\begin{aligned}
\mathbf{F} &= \begin{bmatrix} \sum_{k=0}^{N-1} f_k \\ \sum_{k=0}^{N-1} f_k e^{-\frac{2\pi i k}{N}} \\ \sum_{k=0}^{N-1} f_k e^{-\frac{4\pi i k}{N}} \\ \vdots \\ \sum_{k=0}^{N-1} f_k e^{-\frac{2\pi i (N-1)k}{N}} \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & e^{-\frac{2\pi i}{N}} & e^{-\frac{4\pi i}{N}} & \dots & e^{-\frac{2\pi i (N-1)}{N}} \\ 1 & e^{-\frac{4\pi i}{N}} & e^{-\frac{8\pi i}{N}} & \dots & e^{-\frac{4\pi i (N-1)}{N}} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & e^{-\frac{2\pi i (N-1)}{N}} & e^{-\frac{4\pi i (N-1)}{N}} & \dots & e^{-\frac{2\pi i (N-1)^2}{N}} \end{bmatrix} \mathbf{f}
\end{aligned}$$

This matrix is the DFT matrix. Let $\omega = e^{-\frac{2\pi i}{N}}$. Then, this matrix could be rewritten as:

$$\omega = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \dots & \omega^{(N-1)^2} \end{bmatrix}$$

This is an $N \times N$ matrix. Let us study this matrix for a while. Observe:

$$\omega_{ij} = \omega^{ij}$$

This makes it a symmetric matrix. Further, $\omega^N = 1$.

Let the i^{th} column of the matrix be denoted as ω_i . Then

$$\omega_j^\theta \omega_i = \sum_{k=0}^{N-1} \omega^{(i-j)k}$$

where A^θ is the conjugate transpose of matrix A . The relationship defines the inner product in this case.

$$\omega_j^\theta \omega_i = \begin{cases} 0 & : i \neq j \\ N & : i = j \end{cases}$$

So the columns are orthogonal to each other. (although not orthonormal). We could give a normalising constant of $\frac{1}{\sqrt{N}}$ to make these orthonormal (so that it becomes easy to invert). Anyways if we do not normalise the inverse of the matrix:

$$\omega^{-1} = \frac{\omega^\theta}{N}$$

Recall how the matrix was already a symmetric one. So we do not need to take the transpose of it. Hence,

$$\omega^{-1} = \frac{\bar{\omega}}{N}$$

This makes the inverse DFT also easier.

3.3 Problems with DFT

Now there are a lot of important caveats with this. The primary issue is that DFT takes a considerable amount of time to execute — on computers. Why? Because there are two loops required to compute the DFT for a single signal. Each loop is of length N . That makes the complexity $O(N^2)$. Also the matrix has N^2 complex numbers. So that makes the space complexity $O(N^2)$. So for a signal, if we have $N \approx 10^7$ and each step is computed in one-trillionth of a second (i.e., 10^{-9} s), the time taken to the DFT of that signal is more than two days. Consider a complex number: 8 bytes are taken for a double data type in standard C. A complex number usually has two doubles so defining that way, a complex number will be 16 bytes. Then the total storage required to store the matrix is 1.6PB. That is the same as stacking 400 external hard disks, each of 4TB memory to store a single matrix.

There is however an interesting work around for the space complexity. We can use the periodicity of the complex number ω . Let us show how. Instead of using the matrix formula of the DFT,

$$F = \omega f$$

we can keep the summations. Now we build an array W of length N such that

$$W[i] = \omega^i$$

. Observe that since $\omega^N = 1$, we can say

$$\omega^m = \omega^{m \bmod N} = W[m]$$

This array has space complexity $O(N)$, that is 160 MB. Of course this too is a big amount of storage space. However, I won't compute the DFT of some million points on a local device, I would almost always use a competent machine which is designed for these purposes. And 160MB is one out of 10 million parts of the previous amount.

The time complexity issue however remains, that is, if we do not exploit the symmetry of DFT to optimise our steps. Enter FFT.

3.4 The Fast Fourier Transform

The FFT is not a clever approximation of the DFT. Instead it is a clever observation of the redundant computations in DFT. By reorganizing these computations, large DFTs can be expressed in terms of several smaller DFTs. This recursive decomposition is what gives the FFT its remarkable efficiency. The most widely used FFT is the Cooley–Tukey algorithm, which repeatedly divides the problem into two smaller problems: known as the radix-2 FFT. The implementation discussed here performs FFT on signals that that $N = 2^p$ for some integer p .

So, if the signal points are x_n and $\omega_N = e^{-\frac{2\pi i}{N}}$ then the DFT of it is given by

$$X_k = \sum_{n=0}^{N-1} x_n \omega_N^{kn}$$

Here ω_N is the twiddle factor. Now we can separate into even and odd indices for $n = 2m$ and $n = 2m + 1$. Then,

$$X_k = \sum_{m=0}^{\frac{N}{2}-1} x_{2m} \omega_N^{2mk} + \sum_{m=0}^{\frac{N}{2}-1} x_{2m+1} \omega_N^{(2m+1)k}$$

Now, look at this:

$$\begin{aligned} \omega_N^{2u} &= e^{-\frac{4\pi i u}{N}} = e^{-\frac{2\pi i u}{\frac{N}{2}}} = \omega_{\frac{N}{2}}^u \\ \omega_N^{u+\frac{N}{2}} &= e^{-\frac{2\pi i (u+\frac{N}{2})}{N}} = e^{-\frac{2\pi i u}{N}} e^{-\pi i} = -\omega_N^u \end{aligned}$$

And so we have

$$\begin{aligned} X_k &= \sum_{m=0}^{\frac{N}{2}-1} x_{2m} \omega_N^{2mk} + \sum_{m=0}^{\frac{N}{2}-1} x_{2m+1} \omega_N^{(2m+1)k} \\ &= \sum_{m=0}^{\frac{N}{2}-1} x_{2m} \omega_{\frac{N}{2}}^{mk} + \omega_N^k \sum_{m=0}^{\frac{N}{2}-1} x_{2m+1} \omega_{\frac{N}{2}}^{mk} \\ &= E_k + \omega_N^k O_k \\ X_{k+\frac{N}{2}} &= E_k - \omega_N^k O_k \end{aligned}$$

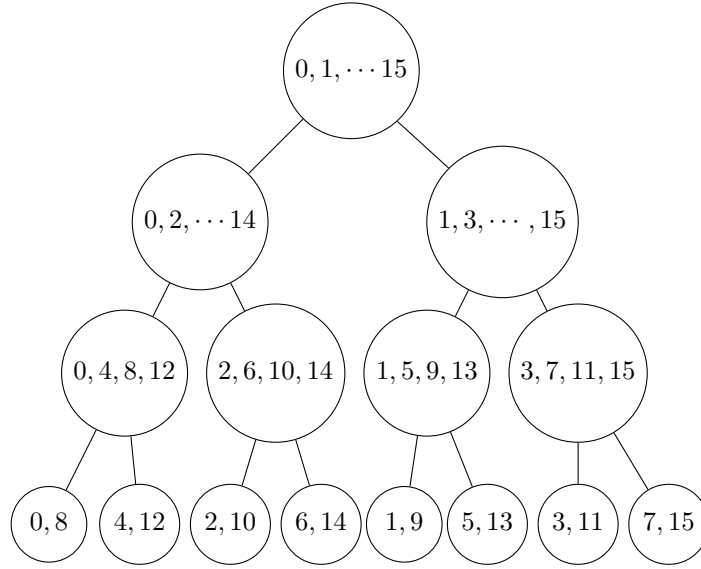
This is a recursive formula. This can be arranged into a binary tree, with the height of the tree being roughly $O(\log N)$ and $O(N)$ being the computation in each step. Making the total time complexity (or more accurately the computational cost) is $O(N \log N)$. This is a lot better than $O(N^2)$ in DFT. Let us consider the case of $N = 16$. So,

$$X_k = \sum_{n=0}^{15} x_n \omega_{16}^{kn}$$

where $\omega_{16}^{kn} = e^{-\frac{2\pi i kn}{16}}$. Then,

$$\begin{aligned} X_k &= E_k + \omega_{16}^k O_k \\ E_k &= \sum_{n=0}^7 x_{2n} \omega_8^{kn} \\ O_k &= \sum_{n=0}^7 x_{2n+1} \omega_8^{kn} \end{aligned}$$

Further, this recursion continues. We could visualise this by the tree diagram:



This is the tree diagram of what happens.

How do we take the inverse FFT? We use the complex conjugate of the twiddle factor and divide the terms by N just like we did for inverse DFT.

We now move on to spectral derivatives, an application of everything we've seen till now. We have spent a lot of time around building the math that is required. And now, we shall see it in action!

Let's look at some complex number magic first, something we all know.

4 Spectral Methods

4.1 Multiplication, Rotation and Differentiation — All At Once!

Consider a complex number z with unit modulus, i.e., $|z| = 1$. Such a complex number is supposed to lie on a unit circle centered at 0 in the Complex plane. Then, we could represent it as $z = \cos \theta + i \sin \theta$ or equivalently $z = e^{i\theta}$, where θ is the angle between z and the real axis. The angle θ is also called the phase of the complex number or its argument.

What is the rate of change of z with respect to the angle θ . Well we could do an easy computation for this:

$$\frac{dz}{d\theta} = \frac{d}{d\theta} e^{i\theta} = i e^{i\theta}$$

This on one hand confirms the known fact that for the differentiation operator in θ , the eigenfunction is $e^{k\theta}$ for any complex number k . But on the other hand see what has happened! Multiplication by i means a pure rotation of 90° in the Complex plane. Why? Because:

$$i = e^{i\pi/2}$$

And so,

$$\frac{d}{d\theta} e^{i\theta} = e^{i(\theta + \frac{\pi}{2})}$$

So, I could multiply z by i , rotate z through 90° or differentiate z and the result is still the same. For simplicity we shall be writing $\frac{d}{d\theta}$ as D_θ .

More generally,

$$\begin{aligned} D_\theta e^{i\alpha\theta} &= i\alpha e^{i\alpha\theta} \\ D_\theta e^{if(\theta)} &= i f'(\theta) e^{if(\theta)} \end{aligned}$$

We could go for higher order derivatives to prove:

$$D_\theta^\beta e^{i\alpha\theta} = (i\alpha)^\beta e^{i\alpha\theta}$$

for whole number β . We will not go to the discussion of fractional derivatives in this main text.

4.2 Spectral Derivatives

Not much is left to guess how we are going to unfold this topic. Let us have a signal, or a function $f(x)$. The Fourier transform of this is

$$\hat{f}(s) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i s x} dx$$

as we all know by now. So, we do the following: Let us also have a function or a signal $g(x)$, which is defined such that

$$\begin{aligned}\hat{g}(s) &= \int_{-\infty}^{\infty} g(x) e^{-2\pi i s x} dx \\ \hat{g}(s) &= 2\pi i s \hat{f}(s)\end{aligned}$$

I want to take the Inverse Fourier transform of $\hat{g}(s)$ to know the relationship between f and g .

$$\begin{aligned}D_x f(x) &= D_x \left(\int_{-\infty}^{\infty} \hat{f}(s) e^{2\pi i s x} ds \right) \\ &= \int_{-\infty}^{\infty} \hat{f}(s) (2\pi i s e^{2\pi i s x}) ds \\ &= \int_{-\infty}^{\infty} \hat{g}(s) e^{2\pi i s x} ds \\ &= g(x)\end{aligned}$$

So without beating around the bush, we state the algorithm very clearly:

1. Take the Fourier Transform of a signal (Use FFT)
2. You get the frequency spectrum. Multiply the frequency spectrum with the respective angular frequencies and i (Multiply by $2\pi i s$)
3. Now do inverse Fourier transform of the new spectrum (Use Inverse FFT)

This is how you'd get the differentiated signal. Now, why is this important? A lot of times you'd need to take derivatives of signals. And quite a lot of times they are in the form of discrete data points, and spectral differentiation works good in those cases. The overall computational complexity of spectral differentiation will be $O(N \log N)$ because the complexity is dominated by the FFT and IFFT complexities: $T(N) + 2T(N \log N) = O(N \log N)$.

What is the error associated with this method? I mean instead of differentiating the conventional way, we are indeed opting for a different process. But doing so are we compromising anything? Let's see: The error will be given as

$$\text{Total Error} = \text{Round off Errors} + \text{Truncation Error in the Series}$$

Let us see what is the truncation error of a Fourier series for a smooth function

$$\begin{aligned}\text{trunc err} &= |f(x) - F_N\{f\}| \\ &= |F_{\infty}\{f\} - F_N\{f\}| \\ &= |F_{>N}\{f\}| \\ &\leq \left| \sum_{k>N} C e^{-\alpha k} \right| \\ &= O(e^{-N})\end{aligned}$$

And the round off error

$$\text{round off err} = \epsilon O(N \log N)$$

Where ϵ is the floating point error. So the total error is

$$\text{Total Error} = O(e^{-\alpha N}) + \epsilon N \log N$$

Which is quite good compared to polynomial errors in case of normal differentiation.

5 Author's Note

This has become much more than just documentation of a project. This project touches upon a lot of topics. Fourier transform has many more utilities and a lot of other authors have written about this topic in their own way. This was my perspective on Fourier Transforms. A lot more could be indeed discussed, but I think this is a good point to stop. Do look at the appendix for some really good proofs and identities.

6 References

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7 Appendix

7.1 Prove that $\int_{-\infty}^{\infty} \frac{\sin t}{t} dt = \pi$

Proof 1. Using Fourier Transform.

Assume we have a signal or function f such that

$$f(x) = \begin{cases} b : |x| \leq a \\ 0 : |x| > a \end{cases}$$

where $b \in \mathbb{R} - \{0\}$ and $a > 0$. So, the Fourier transform of this function:

$$\begin{aligned} \hat{f}(s) &= \int_{-\infty}^{\infty} f(x) e^{2\pi i s x} dx \\ &= b \int_{-a}^a e^{2\pi i s x} dx \\ &= b \frac{e^{2\pi i s a} - e^{-2\pi i s a}}{2\pi i s} \\ &= b \frac{\sin(2\pi s a)}{\pi s} \end{aligned}$$

Now,

$$\begin{aligned}
f(x) &= \int_{-\infty}^{\infty} \hat{f}(s) e^{2\pi i s x} ds \\
&= b \int_{-\infty}^{\infty} \frac{\sin(2\pi s a)}{\pi s} e^{2\pi i s x} ds \\
&= 2ab \int_{-\infty}^{\infty} \frac{\sin(2\pi s a)}{2\pi s a} e^{2\pi i s x} ds \\
f(0) &= 2ab \int_{-\infty}^{\infty} \frac{\sin(2\pi s a)}{2\pi s a} ds \\
b &= \frac{b}{\pi} \int_{-\infty}^{\infty} \frac{\sin t}{t} dt
\end{aligned}$$

So,

$$\int_{-\infty}^{\infty} \frac{\sin t}{t} dt = \pi$$

This concludes the first proof.

Proof 2. Using Feynman's Technique

Let us define

$$I(\alpha) = \int_0^{\infty} \frac{\sin x}{x} e^{-\alpha x} dx.$$

Differentiating under the integral sign,

$$\begin{aligned}
I'(\alpha) &= \int_0^{\infty} \frac{\partial}{\partial \alpha} \left(\frac{\sin x}{x} e^{-\alpha x} \right) dx \\
&= - \int_0^{\infty} e^{-\alpha x} \sin x dx.
\end{aligned}$$

Now let

$$J = \int_0^{\infty} e^{-\alpha x} \sin x dx, \quad K = \int_0^{\infty} e^{-\alpha x} \cos x dx.$$

Integrating by parts,

$$\begin{aligned}
J &= \left[-\frac{e^{-\alpha x} \sin x}{\alpha} \right]_0^{\infty} + \frac{1}{\alpha} \int_0^{\infty} e^{-\alpha x} \cos x dx \\
&= \frac{K}{\alpha},
\end{aligned}$$

and similarly,

$$\begin{aligned}
K &= \left[-\frac{e^{-\alpha x} \cos x}{\alpha} \right]_0^{\infty} - \frac{1}{\alpha} \int_0^{\infty} e^{-\alpha x} \sin x dx \\
&= \frac{1}{\alpha} - \frac{J}{\alpha}.
\end{aligned}$$

Substituting the expression for K into the equation for J ,

$$J = \frac{1}{\alpha} \left(\frac{1}{\alpha} - \frac{J}{\alpha} \right),$$

which gives

$$J \left(1 + \frac{1}{\alpha^2} \right) = \frac{1}{\alpha^2},$$

and hence

$$J = \frac{1}{1 + \alpha^2}.$$

Therefore,

$$I'(\alpha) = -\frac{1}{1 + \alpha^2}.$$

Integrating with respect to α ,

$$I(\alpha) = -\arctan(\alpha) + C.$$

Since

$$\lim_{\alpha \rightarrow \infty} I(\alpha) = 0,$$

we obtain

$$0 = -\frac{\pi}{2} + C,$$

so

$$C = \frac{\pi}{2}.$$

Thus,

$$I(\alpha) = \frac{\pi}{2} - \arctan(\alpha).$$

Finally, setting $\alpha = 0$,

$$\int_0^\infty \frac{\sin x}{x} dx = I(0) = \frac{\pi}{2}.$$

Hence,

$$\int_{-\infty}^\infty \frac{\sin x}{x} dx = 2 \int_0^\infty \frac{\sin x}{x} dx = \pi$$

7.2 Cauchy-Schwarz Inequality

Let us have the vector space R^n . Now, let $u, v \in R^n$ such that $u = (u_1, u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n)$. Then, the inner product is the dot product. And the angle θ is related as

$$\begin{aligned}\|u\|\|v\|\cos\theta &= \langle u, v \rangle \\ \cos\theta &= \frac{\sum_{k=1}^n u_k v_k}{(\sum_{k=1}^n u_k^2)^{\frac{1}{2}} (\sum_{k=1}^n v_k^2)^{\frac{1}{2}}} \\ \cos^2\theta &= \frac{(\sum_{k=1}^n u_k v_k)^2}{(\sum_{k=1}^n u_k^2)(\sum_{k=1}^n v_k^2)}\end{aligned}$$

Since, $0 \leq \cos^2\theta \leq 1$, we have

$$\left(\sum_{k=1}^n u_k v_k\right)^2 \leq \left(\sum_{k=1}^n u_k^2\right) \left(\sum_{k=1}^n v_k^2\right)$$

Equality holds when $u_k = \lambda v_k$ for some real λ , or when the vectors are perfectly aligned in the same or opposite direction.

7.3 Proof of Euler's Formula

Due to Taylor's theorem, we know:

$$\begin{aligned}e^{ax} &= \sum_{k=0}^{\infty} \frac{(ax)^k}{k!} \\ \sin(x) &= \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!} (-1)^k \\ \cos(x) &= \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!} (-1)^k\end{aligned}$$

An intelligent recombination of the terms by putting $a = i = \sqrt{-1}$

$$\begin{aligned}e^{ix} &= \sum_{k=0}^{\infty} \frac{(ix)^k}{k!} \\ &= \sum_{n=0}^{\infty} \frac{(ix)^{2n}}{(2n)!} + \sum_{n=0}^{\infty} \frac{(ix)^{2n+1}}{(2n+1)!} \\ &= \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} (-1)^n + i \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} (-1)^n \\ &= \cos(x) + i \sin(x)\end{aligned}$$

We could also prove in another way. Let $z(x) = \cos(x) + i \sin(x)$. So, $z(0) = 1$.

$$\begin{aligned}z'(x) &= iz(x) \\ \ln z(x) &= ix + C \\ \ln z(0) &= C \\ 0 &= C \\ \ln z(x) &= ix \\ z(x) &= e^{ix}\end{aligned}$$

Although I believe the first proof is much more substantial.

7.4 An Attempt at Fractional Derivatives

When attempting to understand spectral derivatives we came across an equation:

$$D_x^\beta e^{i\alpha x} = (i\alpha)^\beta e^{i\alpha x}$$

We can proof by induction that this holds for whole numbers and even integers. But is it true for all real β ? In general this is the question of fractional derivatives. For example for $\beta = \frac{1}{2}$, does this formula still hold true? or does it break down?

We could do an experiment. Suppose we define an operator S such that $S^2 f(x) = f'(x)$, then clearly $S^2 = D_x$, which means $S = D_x^{\frac{1}{2}}$. Our method clearly does that! However the important caveat here is our method will work only for those functions that live in $L^2(R)$. The reason should be clear by now. Because only functions living in $L^2(R)$ have clearly defined Fourier transforms and otherwise we'd have to do numerical approximation.

I guess the last statement sums up the entire document.

* * *